



Fig. 4: Author's prototype

program. Timer0 and Timer1 are used for display multiplexing and keyboard scanning, respectively. The 'Reset' key is a hardware key and is independent of the software. All other keys are scanned by the source program. The main task in this software development is the time division multiplexing for displaying a particular number on the seven-segment display unit. The keyboard scanning is comparatively

easy. The functions 'display()' and 'display_dp()' are used to display the number on the eight seven-segment display digits. The display function displays negative sign when the code 11H is recognised and it remains blank when the code 10H is recognised in the software. If any numerical key from '0' to '9' is pressed the display shifts the displayed number to left by one digit and the new digit is displayed as the right-most digit. When '00' key is pressed the displayed number shifts by two digits and 00 is displayed as the two right-most digits. If the keys '+', '-', '*', '/', or 'X' are pressed the displayed number will be stored as the first number and the next entered number will be stored as the second number. When the keys '1/X', '√X'

(square root) or '=' are pressed the answer will be displayed. The '+/-' key will display either '-' (negative) or blank (for positive) as the left-most digit. The back space key '←' will cancel the last pressed digit. The cancel key 'C' will cancel the whole entered number.

Limitations

1. The calculator can calculate accurately up to seven digits. If the answer is more than that of seven digits (e.g., more than 9999999) then the displayed answer may be correct or near to correct.

2. Sometimes we expect an integer answer like 125, but the display shows 125.00001. It means that the embedded C gives approximately correct answer for some calculations.

EFY note. The source codes of this article are available on www.efymag.com website. ●

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